

Juan José Aguado

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EDUCATION

Pontificia Universidad Javeriana <i>Systems and Computer Engineering (Computer Science)</i>	Cali, VA Aug. 2020 – Jun. 2025
Universidad Icesi <i>Master in Applied Artificial Intelligence</i>	Cali, VA Jul. 2025 – Dec. 2026

EXPERIENCE

Tasker - Generative AI Models <i>Outlier</i>	Jun. 2024 – Present <i>Remote</i>
- Evaluated the performance of generative AI models (text, audio, image) based on criteria such as precision, instruction-following, and safety. - Conducted evaluations in English, ensuring a high standard of language proficiency and communication in reports. - Provided feedback on model's outputs, helping improve their accuracy and compliance with security protocols.	
Database Administrator <i>Fundación Casa de Colombia</i>	Jan. 2024 – Jun. 2024 <i>Cali, VA</i>
- Spearheaded the design, creation, and deployment of a comprehensive database for a software solution aimed at streamlining operations at Casa de Colombia. - Developed and maintained PostgreSQL databases to efficiently manage and store crucial information, supporting the provision of essential health and education services to children. - Utilized Jira for project management and issue tracking, ensuring efficient workflow and task management. - Employed Git and Bitbucket for version control and collaborative development, maintaining a robust and organized codebase.	
Research Assistant - Applied Software Engineering Research Group <i>Pontificia Universidad Javeriana</i>	Jun. 2024 – Present <i>Cali, Colombia</i>
- Working on the university's automatic email sending service. Developed and modified functionalities using Java and Maven. - Utilized Postman to test and validate API endpoints for automated email delivery. - Collaborated with teams to enhance email service performance and maintainable code structure. - Integrated solutions and performed troubleshooting on email service errors and logs to improve user experience.	
Data Scientist <i>Atinna</i>	Oct. 2024 – Present <i>Cali, Colombia</i>
- Developing social listening systems by consuming and transforming large-scale datasets using Python, Git, Hugging Face, OpenAI, and LangChain for LLM-based processing and analysis. - Utilizing embeddings for clustering and semantic similarity tasks to identify meaningful relationships and topic structures within multilingual social media data. - Performing fine-tuning experiments through few-shot learning and DSPy, enhancing LLM performance on sentiment, stance, and political alignment classification tasks. - Designing and managing a relational database with Azure SQL to store structured listening outputs and metadata for dynamic Power BI visualizations. - Integrating Langfuse with the backend to monitor and version prompts directly from the Langfuse platform, ensuring traceability and controlled prompt evolution. - Building a backend architecture to automate the complete workflow of social listening pipelines, including ingestion, model execution, and report generation. - Creating interactive visualizations, dynamic maps, and strategic analytics dashboards to support data-driven decisions and client reporting. - Optimizing collaborative Git workflows to enhance productivity, maintain reproducibility, and improve model deployment cycles.	

PROJECTS

UVa Programming Problems | *Python*

January 2023 – June 2023

- Solved numerous programming problems from the UVa Online Judge, a platform hosted by the University of Valladolid with over 4300 problems.
- Utilized various data structures such as arrays, linked lists, trees, queues, and graphs to optimize solutions.
- Implemented algorithms including bisection, binary search, backtracking, dynamic programming, and greedy algorithms to solve complex problems.
- Gained deep understanding of problem-solving techniques and algorithm optimization through continuous practice and refinement of solutions.

Statistical Analysis and Data Science Projects | *R, Markdown*

January 2024

- Conducted comprehensive analysis of the real estate market in Cali, Colombia, using data from B&C real estate agency, focusing on property prices, locations, features, and sales.
- Preprocessed and cleaned real estate data, analyzed descriptive statistics, and created visualizations to communicate findings effectively.
- Performed various statistical simulations, including Monte Carlo Simulation to estimate the value of π , and analyzed bias, efficiency, and consistency of different estimators.
- Verified the Central Limit Theorem through simulations, observed its implications on sample distributions, and constructed confidence intervals using bootstrap methods.
- Built multiple linear regression models to predict real estate prices based on the area, validated model assumptions, performed hypothesis testing, and compared models to find the best fit.
- Developed a deep understanding of statistical analysis and data science techniques using R, enhancing skills in data manipulation, visualization, and modeling.

COURSES

HarvardX CS50 | *Django, HTML, Git, CSS, Javascript, SQL*

April 2024

- Developed a full-stack web application using Django, implementing both the backend with RESTful APIs and the frontend with HTML, CSS, and JavaScript.
- Utilized Git and GitHub for version control and collaboration, ensuring efficient project management and code integrity.
- Implemented interactive features and dynamic content using JavaScript for seamless user experience.
- Designed and managed relational databases with SQL, ensuring data integrity and efficient query handling.
- Applied best practices in web security, testing, and deployment, ensuring robust and reliable applications.

SKILLS

Programming languages: Python, C/C++, SQL (Postgres, Azure SQL), JavaScript, HTML/CSS, R

Frameworks: Django

Libraries: pandas, NumPy, Matplotlib, Seaborn, SciKitLearn, Hugging Face, OpenAI, LangChain, DSPy, Plotly

Developer Tools: Git, Docker, VS Code, PostgreSQL (PgAdmin), Azure SQL Server, Jira, Bitbucket, RStudio, VirtualBox, Google Colab, Eclipse, Maven, Langfuse, Power BI

Cloud Platforms: Azure, AWS

Languages: Spanish (native), English (advanced)